

GreenMart Supermarket

Domain Knowledge Description

GreenMart is a supermarket company that operates multiple branches. Each branch belongs to exactly one supermarket company and has a unique branch code.

GreenMart organizes its stores into five types of departments: Dry Food, Household Goods, Fresh Produce, Beverages, and Personal Care. Every branch must have at least one Dry Food Department and at least one Household Goods Department. The other three types of departments are optional. A branch may have only these five types of departments, and each department belongs to exactly one branch.

Each department must offer at least five different product categories. For example, a Dry Food Department may offer rice, pasta, cereal, canned food, and flour. A Household Goods Department may offer cleaning products, kitchenware, storage containers, laundry accessories, and small household tools.

Products are classified into two main groups: Food Products and Non-Food Products. Food Products include Dry Food Products, Fresh Produce Products, and Beverage Products. Non-Food Products include Household Goods Products and Personal Care Products. Household Cleaning Products are a specific type of Household Goods Product.

Each department can stock only products belonging to its corresponding product group. For example, a Dry Food Department can stock only Dry Food Products, while a Household Goods Department can stock only Household Goods Products. In addition, every Household Goods Department must stock at least one Household Cleaning Product.

In this simplified system, each product is stocked by exactly one department and belongs to exactly one product category. Every product must have exactly one name and exactly one price. The price must be a non-negative decimal number.

Student Tasks

Based on the domain description, design an OWL ontology using Protégé.

1. Identify the required classes and organize them into an appropriate class hierarchy.
2. Define the necessary object properties and data properties, including their domains and ranges.
3. Define class descriptions using appropriate OWL restrictions to represent the requirements in the domain.